AI improves highway safety in Las Vegas

Artificial intelligence (AI) is helping improve safety along a stretch of Las Vegas' busiest highways. Nevada Highway Patrol says that a year-long partnership between public safety agencies and an Israeli startup technology firm, Waycare, has resulted in seventeen per cent reduction in crashes along a portion of north-bound Interstate 15 just west of Las Vegas Strip.

According to *www.usnews.com*, the platform uses in-vehicle information and municipal traffic data to understand road conditions in real time. When an area at high risk for an incident is identified, Waycare alerts traffic agencies when and where to take preventive action. The Regional Transportation Commission of Southern Nevada uses dynamic message boards to relay advanced warning of an incident, alerting drivers to reduce speed and drive cautiously.

Nevada Highway Patrol then deploys its vehicles in high-visibility mode along the highway in conjunction with Nevada Department of Transportation, which assures that safety barriers are in place for the police officers on the highway.

During the program, 91 per cent of drivers travelling at a high speed slowed down in areas where preventive measures were deployed.

"Results of this pilot program are a clear signal that AI and deep learning, when deployed in collaboration with traffic management and enforcement agencies, can have a dramatic impact on improving the safety of even our busiest and most at-risk highways," said Noam Maital, co-founder and chief executive officer, Waycare.

Snapdragon X55 is the world's most advanced commercial 5G modem

Qualcomm's Snapdragon X55 is designed to accelerate global 5G rollout and bring 5G to a broad range of device categories and applications beyond smartphones, including hotspots, Wi-Fi routers, always-connected PCs, laptops, tablets, extended reality devices and connected cars. Combined with Qualcomm RF front-end solutions, Snapdragon X55 is designed to support high-power fixed wireless access (FWA) services and equipment.

Snapdragon X55 is designed to provide comprehensive support for 5G NR TDD and FDD, as well as mmWave and sub-6GHz spectrum. It also supports 5G non-standalone (NSA) and standalone (SA) modes, and 5G/4G spectrum sharing. As a result, devices powered by Snapdragon X55 will offer extensive flexibility to operators for utilising their spectrum assets to help bring the best connected experiences to consumers.

In addition to providing flexibility, 5G/4G spectrum sharing support also helps operators improve network efficiency and capacity. The added full-dimension MIMO (FD-MIMO) support in Snapdragon X55 allows operators to utilise both azimuth (horizontal) and elevation (vertical) beamforming to boost spectral efficiency.

The future of pain relief is here

In an attempt to create an antenna that was not constrained by physical dimensions, inventors from Utah created the technology that would one day become Kailo—a nanotech patch that boosts electrical signals in the body, significantly reducing pain.

Kailo, the world's first nanotech bio antenna (Credit: www.indiegogo.com)

When a user wears Kailo, it interacts with the body's electrical system. Each patented Kailo contains billions of charged nanocapacitors that work as a bio-antenna, assisting the body in clear communication and reducing the signals that cause pain.

There are three layers of Kailo: carrier layer, nanoparticles and substrate. The carrier layer is a non-conductive layer made from a special kind of synthetic polymer that works as a base for the particle mixture. This laminate helps protect the particles from friction and water damage.

The nanoparticles layer interacts with the body's electrical system. It contains billions of charged nanocapacitors that work as an antenna, assisting the body in clear communication and reducing the signals that cause pain.

In the third layer, the nanoparticles are flood-coated with a patented substrate that holds everything in place to create a dust-free and water-tight seal.

Each Kailo comes with a pack of adhesives that can be easily applied to the back of Kailo for seamless application. When the adhesive loses its stick, the user can simply peel it off and apply a new one for next time. Kailo also works with other commonly-available athletic or medical tapes to hold it in place.